

The output of PORTA yields (after 0.11 seconds of computation time) a complete minimal system of inequalities for the convex hull of these points, namely

DIM = 4

VALID

7 49 343 2401

INEQUALITIES_SECTION

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( 1) -210x1+107x2-18x3+x4 <= 0
( 2) -140x1+ 83x2-16x3+x4 <= 0
( 3) - 84x1+ 61x2-14x3+x4 <= 0
( 4) - 42x1+ 41x2-12x3+x4 <= 0
( 5) - 14x1+ 23x2-10x3+x4 <= 0
( 6) + 6x1- 11x2+ 6x3-x4 <= 0
( 7) + 12x1- 19x2+ 8x3-x4 <= 0
( 8) + 20x1- 29x2+10x3-x4 <= 0
( 9) + 30x1- 41x2+12x3-x4 <= 0
(10) + 42x1- 55x2+14x3-x4 <= 0
(11) + 50x1- 35x2+10x3-x4 <= 24
(12) + 78x1- 49x2+12x3-x4 <= 40
(13) +112x1- 65x2+14x3-x4 <= 60
(14) +152x1- 83x2+16x3-x4 <= 84
(15) +154x1- 71x2+14x3-x4 <= 120
(16) +216x1- 91x2+16x3-x4 <= 180
(17) +288x1-113x2+18x3-x4 <= 252
(18) +342x1-119x2+18x3-x4 <= 360
(19) +450x1-145x2+20x3-x4 <= 504
(20) +638x1-179x2+22x3-x4 <= 840
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END

In particular, this polytope has 20 facets.

The “-v” option of the PORTA program produces also the vertex-facet incidence matrix given on the next page, from which we can derive the complete combinatorial structure of the polytope.

In this matrix, the vertex-facet incidences are denoted by *’s. From the matrix we can determine that $C_4(8)$ is simplicial, since every facet has exactly 4 vertices, corresponding to exactly 4 *’s in every row — this is also recorded in the last column. We also see that every vertex is on exactly 10 facets: there are 10 *’s in every column; see the bottom row of the matrix.

From the rows of the matrix we can observe the following pattern, known as *Gale’s evenness condition*: every segment of consecutive *’s is of even length if it is not an initial or a final segment, that is, if it is preceded and followed by a dot. (For this, the vertices of $C_d(n)$ are labeled $1, \dots, n$, with i corresponding to $\mathbf{x}(t_i)$.)

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1		*.... *** :	4
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4		*...* ... :	4
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6		****. ... :	4
7		*** ** ... :	4
8		*** * ... :	4
9		****. *** :	4
10		****. ... :	4
11		****. ... :	4
12		*** * ... :	4
13		****. *** :	4
14		****. ... :	4
15		****. ... :	4
16		****. ... :	4
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19		****. ... :	4
20		****. ... :	4
			
#		11111 111	
		00000 000	

From this pattern, one can derive that any two vertices of the polytope are adjacent. We can also check this directly: every pair of vertices is contained in at least 3 facets. So, the edge 12 is contained in the facets (5) = 1238, (6) = 1234, (7) = 1245, (8) = 1256, (9) = 1267, and (10) = 1278. Similarly, the edge [1, 3] is contained in the facets (4) = 1348, (5) = 1238 and (6) = 1234.

Finally, we can note that there is a combinatorial symmetry that sends vertex i to vertex $9 - i$; see Exercise 0.7.

The following theorem and corollary contain a complete description of the combinatorial structure of the cyclic polytopes — as suggested by our computation. Here we break our promise not to do any proofs in this introduction: mainly because the proofs are fun, and the results are a little surprising (see Corollary 0.8!).

Theorem 0.7 (Gale’s evenness condition). (Gale [221])

Let $n > d \geq 2$. We will use $[n]$ to denote the set $\{1, \dots, n\}$, and choose real parameters $t_1 < t_2 < \dots < t_n$.

The cyclic polytope

$$C_d(n) = \text{conv}\{\mathbf{x}(t_1), \dots, \mathbf{x}(t_n)\}$$

is a simplicial d -polytope. A d -subset $S \subseteq [n]$ forms a facet of $C_d(n)$ if and only if the following “evenness condition” is satisfied:

If $i < j$ are not in S , then the number of $k \in S$ between i and j is even:

$$2 \mid \#\{k : k \in S, i < k < j\} \quad \text{for } i, j \notin S.$$

Proof. Recall the famous Vandermonde determinant identity

$$\det \begin{pmatrix} 1 & 1 & \dots & 1 \\ \mathbf{x}(t_0) & \mathbf{x}(t_1) & \dots & \mathbf{x}(t_d) \end{pmatrix} = \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ t_0 & t_1 & \dots & t_d \\ \vdots & \vdots & \ddots & \vdots \\ t_0^{d-1} & t_1^{d-1} & \dots & t_d^{d-1} \\ t_0^d & t_1^d & \dots & t_d^d \end{pmatrix} = \prod_{0 \leq i < j \leq d} (t_j - t_i).$$

This is easily proved by observing that both sides are polynomials and that the determinants vanish whenever we have $t_i = t_j$ for some $i \neq j$. From the identity, one sees that no $d+1$ points on the moment curve are affinely dependent. In particular, this shows that $C_d(n)$ is a simplicial d -polytope.

Now let $S = \{i_1, \dots, i_d\} \subseteq [n]$. Then the hyperplane H_S through the corresponding points $\mathbf{x}(t_{i_s})$ is given by

$$H_S = \{\mathbf{x} \in \mathbb{R}^d : F_S(\mathbf{x}) = 0\},$$

where

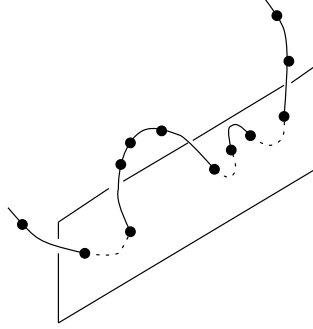
$$F_S(\mathbf{x}) := \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ \mathbf{x} & \mathbf{x}(t_{i_1}) & \dots & \mathbf{x}(t_{i_d}) \end{pmatrix}$$

In fact, $F_S(\mathbf{x})$ is a linear function in $\mathbf{x}$, and it vanishes on the prescribed points.

(The reader should check, at least for one or two examples, that the inequalities that we have computed for $C_4(8)$ have the form “ $\pm F_S(\mathbf{x}) \leq r_S$ ” for some $r_S \in \mathbb{R}$.)

Now we let the point $\mathbf{x}(t)$ move on the moment curve $\{\mathbf{x}(t) : t \in \mathbb{R}\}$. Note that $F_S(\mathbf{x}(t))$ is a polynomial in t of degree d . It vanishes for $t = t_{i_s}$:

thus it has d different zeroes, and changes the sign at each of them. The following sketch is supposed to illustrate this.



Now S forms a facet if and only if $F_S(\mathbf{x}(t_i))$ has the same sign for all the points $\mathbf{x}(t_i)$ with $i \in [n] \setminus S$; that is, if $F_S(\mathbf{x}(t))$ has an even number of sign changes between $t = t_i$ and $t = t_j$, for $i < j$ and $i, j \in [n] \setminus S$. $\square$

In particular, this criterion shows that the combinatorics of $C_d(t_1, \dots, t_n)$ do not depend on the specific choice of the parameters t_i , so $C_d(n)$ is well defined as a combinatorial equivalence class of polytopes.

It is quite easy to extend the evenness condition to a characterization of all the faces of $C_d(n)$. This characterization then also shows the following corollary (Exercise 0.8), for which we give an independent proof.

Corollary 0.8. *The cyclic polytope $C_d(n)$ is $\lfloor \frac{d}{2} \rfloor$ -neighborly, that is, any subset $S \subseteq [n]$ of $|S| \leq \frac{d}{2}$ vertices forms a face.*

Proof. Let $C_d(n) = C_d(t_1, \dots, t_n)$ with $t_1 < \dots < t_n$, and let $T = \{i_1, \dots, i_k\} \subseteq [n]$ have cardinality $k \leq \frac{d}{2}$. Choose some $\varepsilon > 0$ small enough such that $t_i < t_i + \varepsilon < t_{i+1}$ for all $i < n$, and some $M > t_n + \varepsilon$.

Using $\mathbf{x}(M+1), \mathbf{x}(M+2), \dots$ as dummy points “far out there,” we define a linear function $F_T(\mathbf{x})$ as

$$\det \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 \\ \mathbf{x} & \mathbf{x}(t_{i_1}) & \mathbf{x}(t_{i_1} + \varepsilon) & \cdots & \mathbf{x}(t_{i_k}) & \mathbf{x}(t_{i_k} + \varepsilon) & \mathbf{x}(M+1) & \cdots & \mathbf{x}(M+d-2k) \end{pmatrix}.$$

This is a linear function in $\mathbf{x}$, which vanishes on the points $\mathbf{x}(t_i)$ for $i \in T$. If we consider $F_T(\mathbf{x}(t))$, then this is a polynomial in t of degree d , and has d “obvious” distinct zeroes

$$t_{i_1}, t_{i_1} + \varepsilon, \dots, t_{i_k}, t_{i_k} + \varepsilon, M+1, \dots, M+d-2k.$$

There is an even number of zeroes between $t = t_i$ and $t = t_j$ for $i, j \in [n] \setminus T$, because a zero at $t = t_l$ always comes in a pair with a zero at $t = t_l + \varepsilon$. Thus $F_T(\mathbf{x})$ has the same sign on all the points $\mathbf{x}(t_i) : i \in [n] \setminus T$. $\square$

For $d \leq 3$ Corollary 0.8 just says that the points $\mathbf{x}(t_i)$ form vertices of $C_d(n)$: the points on the moment curve are in convex position. However,

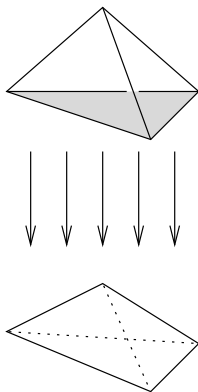
for $d \geq 4$ Corollary 0.8 yields something “counterintuitive”: it describes a property that does not manifest itself in $d \leq 3$ dimensions. Namely, for $d \geq 4$ the polytope $C_d(n)$ has n pairwise adjacent vertices, where n may be much larger than d .

More generally, one defines a d -polytope to be k -neighborly if any subset of k or less vertices is the vertex set of a face of P . In Exercise 0.10, we see that, except for simplices, no polytope is more than $\lfloor \frac{d}{2} \rfloor$ -neighborly. Therefore, polytopes that are $\lfloor \frac{d}{2} \rfloor$ -neighborly are known as *neighborly polytopes*. Thus, by Corollary 0.8, cyclic polytopes are neighborly.

The neighborly polytopes are the solution of various extremal properties. This is one reason why they are important. For example, the famous *upper bound theorem* of McMullen (which we will state and prove in Section 8.4) implies that among all d -polytopes with n vertices, the neighborly ones have the greatest number of facets. In particular, no d -polytope with n vertices has more facets than the cyclic polytope $C_n(d)$.

Example 0.9. If we apply an affine map π to a polytope P , then we get a new polytope $\pi(P)$: this is quite obvious from the definition of a $\mathcal{V}$ -polytope in Definition 0.1. If the affine map is injective, then the image polytope $\pi(P)$ is (affinely) isomorphic to the original one — nothing interesting has happened.

However, one can also take affine maps that project P to a polytope $f(P)$ of lower dimension.



In particular, the convex hull

$$\text{conv}\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{R}^d$$

can be interpreted as the image of the standard simplex $\Delta_{n-1} \subseteq \mathbb{R}^n$, under the linear map $\pi : \mathbb{R}^n \longrightarrow \mathbb{R}^d$, mapping $\mathbf{e}_i \longmapsto \mathbf{x}_i$. This is usually interpreted geometrically as a *projection* of polytopes (which suggests some special choice of coordinates, where $\mathbb{R}^d$ is embedded as a subspace of $\mathbb{R}^n$).

We conclude that a ($\mathcal{V}$ -)polytope is the same thing as the projection of a simplex, and that every projection of a polytope is a polytope as well.

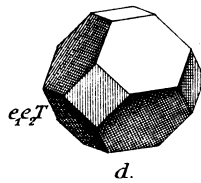
A polytope $P \subseteq \mathbb{R}^d$ is *centrally symmetric* if it has a *center*: a point $\mathbf{x}_0 \in \mathbb{R}^d$ such that $\mathbf{x}_0 + \mathbf{x} \in P$ holds if and only if $\mathbf{x}_0 - \mathbf{x} \in P$. Every affine image (projection) of a crosspolytope is centrally symmetric: if $P = \{A\mathbf{x} + \mathbf{x}_0 : \mathbf{x} \in C_d^\Delta\}$, then P is centrally symmetric with respect to $\mathbf{x}_0$. In fact, every centrally symmetric polytope is the projection of a crosspolytope (Exercise 0.2).

The projections of cubes, called *zonotopes*, form an especially interesting class of polytopes. For example, they encode the structure of linear hyperplane arrangements; see Lecture 7.

Example 0.10. The *permutahedron* $\Pi_{d-1} \subseteq \mathbb{R}^d$ is defined as the convex hull of all vectors that are obtained by permuting the coordinates of the

vector $\begin{pmatrix} 1 \\ 2 \\ \vdots \\ d \end{pmatrix}$. It was apparently first investigated by Schoute [481] in 1911:

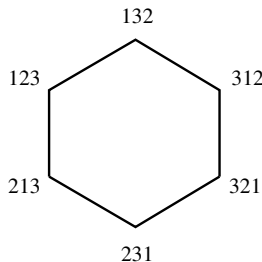
we have taken the following drawing from his paper [481, Fig. 4].



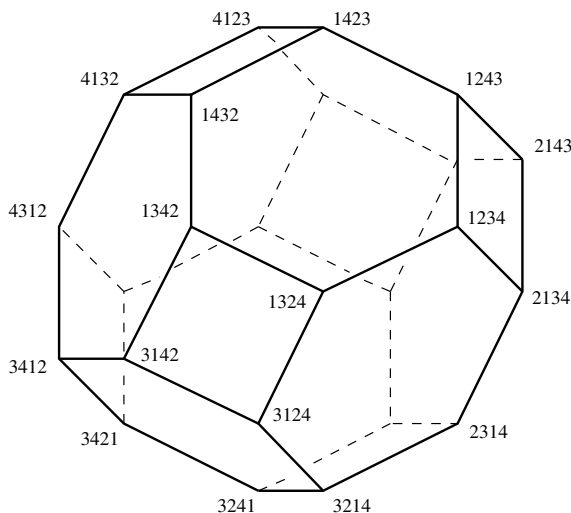
The permutahedron is a very interesting polytope. In fact, it is a simple zonotope (Exercise 0.3), which is rare. Its vertices can be identified with the

permutations in $\mathcal{S}_d$ (namely, by associating with $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}$ the permutation

that maps $x_i \mapsto i$) in such a way that two vertices are connected by an edge if and only if the corresponding permutations differ by an adjacent transposition. Check this in our drawing of Π_2 :



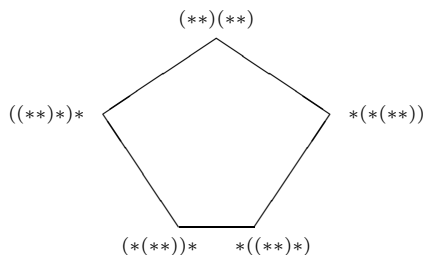
and that of Π_3 :



There is a simple combinatorial description of all the faces of Π_{d-1} : its k -faces correspond to ordered partitions of the set $[d]$ into $d - k$ nonempty parts. Thus the vertices are permutations, and the facets are partitions of $[d]$ into parts $(S, [d] \setminus S)$ with $\emptyset \subset S \subset [d]$.

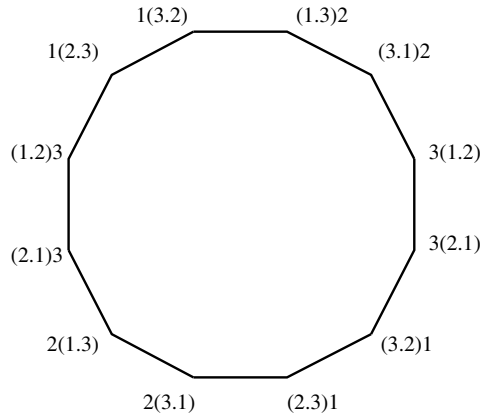
The permutahedron is a classical object; see [96, Example 2.2.5] for further references. We'll meet it again as a zonotope in Section 7.3, and as a fiber polytope (the monotone path polytope of the cube) in Section 9.2.

There is a much more recent counterpart, the *associahedron* K_{n-2} , first described as a combinatorial object by Stasheff [522] in 1964, and constructed as a convex polytope by John Milnor (unpublished, unrecorded), by Mark Haiman [266], and by Carl Lee [355]. The vertices of this (simple) polytope correspond to all the $\frac{1}{n} \binom{2n-2}{n-1}$ different ways of bracketing a string of n -letters, that is, of multiplying an expression $a_1 a_2 \dots a_n$ when multiplication is not associative. Two vertices are adjacent if they correspond to a single application of the associative law. Our figure depicts the 5-gon, which we get as K_{n-2} for $n = 4$:



Whereas the first constructions of the associahedra were very much “ad hoc,” in Lecture 9 we will get an associahedron from a very natural construction due to Gel’fand, Zelevinsky & Kapranov [231, 232]: as the “secondary polytope” of the n -gon [231, Rem. 7c)]. More generally, we will construct “fiber polytopes” there, a concept due to Billera & Sturmfels [78, 79].

Recently, Mikhail M. Kapranov [313] constructed a new combinatorial object $K\Pi_{n-1}$, the *permuto-associahedron*, which combines the permutahedron and the associahedron. (Kapranov denotes it “ KP_n ”.) Its vertices correspond to the different ways of multiplying n terms $a_1, a_2, \dots, a_n$ in arbitrary order, assuming that multiplication is neither commutative nor associative — and again there is a natural way to describe all the faces. Our drawing shows $K\Pi_2$, a 12-gon.



Kapranov [313] showed that the combinatorially defined object $K\Pi_{n-1}$ can for every $n \geq 2$ be realized by a cell complex that is a topological ball. The question of whether the permuto-associahedron (or “Kapranotope”) can be realized as a convex polytope was answered in joint work with Vic Reiner [453] while I was first giving this course; see Section 9.3.

Example 0.11. A class of very interesting polytopes appears in combinatorial optimization: *0/1-polytopes*, all of whose vertex coordinates are 0 or 1 (cf. Schrijver [484], and Ziegler [577]). In other words, a 0/1-polytope is the convex hull of a subset of the vertices of a (unit) cube.

Note that the $(d-1)$ -simplex $\Delta_{d-1} \subseteq \mathbb{R}^d$ is a 0/1-polytope. Similarly, one can study the *hypersimplex* $\Delta_{d-1}(k)$ in $\mathbb{R}^d$, by

$$\begin{aligned} \Delta_{d-1}(k) &= \operatorname{conv}\{\mathbf{v} \in \{0, 1\}^d : \sum_{i=1}^d v_i = k\} \\ &= \{\mathbf{x} \in \mathbb{R}^d : 0 \leq x_i \leq 1 \text{ for } 1 \leq i \leq d, \sum_{i=1}^d x_i = k\} \end{aligned}$$

for $1 \leq k \leq d-1$.

This family includes the standard simplex as $\Delta_{d-1} = \Delta_{d-1}(1)$. The hypersimplex $\Delta_{d-1}(k)$ has $\binom{d}{k}$ vertices, and $2d$ facets, if $2 \leq k \leq d-2$ (but only d facets for $k=1$ or $k=d-1$). For example, the 3-dimensional hypersimplex $\Delta_3(2) \subseteq \mathbb{R}^4$ is combinatorially equivalent to an octahedron.

It seems that the hypersimplices first appeared in Gabriélov, Gel'fand & Losik [218, Sect. 1.6] — in the theory of characteristic classes. See also Gel'fand, Goresky, MacPherson & Serganova [229], and Exercise 5.3(i). These interesting polytopes certainly deserve more study!

Example 0.12. A very “classical” class of 0/1-polytopes (introduced by Birkhoff [83] in 1946) arises from the following construction. Let $\mathcal{S}_d$ denote the set of all permutations of the set $[d]$. With every permutation σ in $\mathcal{S}_d$, we associate the matrix X^σ , given by

$$X_{ij}^\sigma := \begin{cases} 1 & \text{if } \sigma(i) = j, \\ 0 & \text{otherwise.} \end{cases}$$

The matrices X^σ are the 0/1-matrices with exactly one 1 per row and per column. If we identify $\mathbb{R}^{d^2}$ with the set of all real $(d \times d)$ -matrices, then the matrices X^σ are 0/1-vectors in $\mathbb{R}^{d \times d}$, and their convex hull forms a 0/1-polytope

$$P(d) := \text{conv}\{X^\sigma : \sigma \in \mathcal{S}_d\} \subseteq \mathbb{R}^{d^2}.$$

This is an interesting polytope with many names: the *Birkhoff polytope*, the *perfect matching polytope* of $K_{d,d}$, the *assignment polytope*, the *polytope of doubly stochastic matrices*, and so forth.

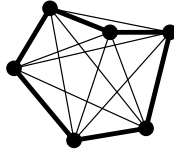
The polytope $P(d)$ has $d!$ vertices (by construction), d^2 facets, and dimension $(d-1)^2$. In fact, a complete linear description is given by

$$P(d) = \left\{ X \in \mathbb{R}^{d \times d} : \begin{aligned} & x_{ij} \geq 0, \text{ for } 1 \leq i, j \leq d, \\ & \sum_{k=1}^d x_{ik} = 1 \text{ for } 1 \leq i \leq d, \\ & \sum_{k=1}^d x_{kj} = 1 \text{ for } 1 \leq j \leq d. \end{aligned} \right\}$$

This is not hard to prove (just do it!): it is a classical result due to Birkhoff [83] and von Neumann [421] independently. (See the treatment by Lovász & Plummer [369].) With this the Birkhoff polytopes are “well described” — that is, we know all the vertices and all the facets. Among many other interesting properties, we note here that $P(d)$ has a canonical center point, given by $x_{ij} = \frac{1}{d}$ for all i and j .

Brualdi and Gibson undertook a detailed study of the Birkhoff polytopes in a series of four papers [136]. Still, there are questions left.

Example 0.13. For a class of nastier 0/1-polytopes, consider the famous *traveling salesman problem* [350], which asks for the shortest possible tour through a complete graph K_n on n vertices, where every edge has a length given. For example, in the graph drawn here ($n = 6$), the length is given by Euclidean distance, and the shortest tour is shown in thick lines.



Every traveling salesman tour can be considered as a subset of n edges, $T \subseteq E(K_n)$, of the graph. We associate with every tour T its “characteristic vector” $\chi_T \in \{0, 1\}^{\binom{n}{2}} \subseteq \mathbb{R}^{\binom{n}{2}}$, that is, the 0/1-vector whose entries indicate which edges are in T , and which are not. Now the *traveling salesman polytope* $Q_T(n)$ is defined as

$$Q_T(n) := \operatorname{conv} \left\{ \chi_T \in \{0, 1\}^{\binom{n}{2}} : \chi_T \text{ is a tour through } K_n \right\}.$$

It is not hard to see that $Q_T(n)$ is a polytope of dimension $\binom{n}{2} - n = n(n-3)/2$. We know the vertices of $Q_T(n)$: they are the $(n-1)!/2$ different Hamilton tours through K_n . Now the question for the shortest tour is answered if we find a vertex that minimizes a linear function: thus the traveling salesman problem is a linear programming problem over $Q_T(n)$.

Similarly, one can define the polytopes $Q'_T(n)$ corresponding to the *asymmetric traveling salesman problem*, which seeks to find the shortest possible directed tour through a complete directed graph K'_n on n vertices, where each of the $n(n-1)$ arcs has a given length. The corresponding polytope $Q'_T(n) \subseteq \mathbb{R}^{n^2-n}$ has dimension $n^2 - 3n + 1$ (for $n \geq 3$), and $(n-1)!$ vertices.

To illustrate that these polytopes are nasty, we just mention the recent result of Billera & Sarangarajan [76] that *every* 0/1-polytope is isomorphic to a face of $Q'_T(n)$, for large enough n . A little trick of Karp [317] [295] shows that (an isomorphic copy of) the asymmetric travelling salesman polytope $Q'_T(n)$ appears as a face of the symmetric travelling salesman polytope $Q_T(2n)$. Thus, the result of Billera & Sarangarajan [76] also applies to the symmetric TSP polytope.

Using linear programming, we could solve the traveling salesman problems efficiently, if we could deal with two major obstacles: we do not know the facet-defining inequalities of $Q_T(n)$, respectively of $Q'_T(n)$, and there are simply too many of them.

In the next lecture we will describe a general method for finding the facets of a polytope given in the form $Q = \operatorname{conv}(V)$. It is the method that makes the PORTA program work. It has successfully been applied to get complete descriptions of the traveling salesman polytopes up to $Q_T(8)$ and

$Q'_T(6)$; see Exercises 0.14 and 1.1(iv). However, it seems that the method does not go beyond that: in general the algorithmic determination of all the facets of Q is certainly much harder and more strenuous than examining all the vertices of Q .

The problem of finding some of the facets, by using the combinatorial properties of the traveling salesman problem, is a central problem for a whole branch of mathematics, called “polyhedral combinatorics” — see Grötschel & Padberg [247] and Jünger, Reinelt & Rinaldi [295] for solid introductions, including detailed information about the structure of the polytopes $Q_T(n)$ and $Q'_T(n)$.

Notes

The principal historical “classics” in the theory of polytopes are the 1852 treatment by Schläfli [473] published in 1901, the books by Brückner [138] (1900), Schoute [480] (1905), and Sommerville [506] (1929), and the volume by Steinitz & Rademacher [527] (1934) about 3-dimensional polytopes. (A very helpful bibliography is Sommerville [507].) The modern theory of polytopes was established by Grünbaum’s 1967 book [252]. It should be stressed that not only did Grünbaum present the major part of what was known at the time, but his book also contains various pieces of progress and substantial original contributions, and has been an inspiring source of problems, ideas, and references to everyone working on polytopes since then.

There are more recent books and surveys on polytopes. Many of them concentrate on aspects related to the upper and lower bound theorems and the g -theorem (among them McMullen & Shephard [403], Brøndsted [133], Stanley [515], and Hibi [274]) and on the various methods of f -vector theory; see Lecture 8. Other aspects are treated in Barnette’s exposition on 3-polytopes [45], Schrijver’s book on optimization [484], and the handbook chapters by Kleinschmidt & Klee [329], and Bayer & Lee [63]. Also, the reader might find Pach’s volume [431] inspiring.

In our lectures we avoid any larger discussion of general convex sets and bodies, as well as of most of the convex-geometric aspects of polytopes. We refer to Bonnesen & Fenchel [124], Schneider [476], and Ewald [201], — the point is that for a convex polytope, we can describe and discuss everything in terms of vertices, edges, facets, etc. (i.e., a finite collection of combinatorial data) and bypass the apparatus of support functionals, nearest point maps, distances, volume, and integration, etc. Correspondingly, in this book we disregard all metric properties of polytopes, such as volume, surface area, and width, which are part of a very interesting theory of their own.